

Project Proposal

Computational Economics





Context and Motivation

- Health risks linked to higher mortality based on lifestyle choices could be mapped and highlighted to help people live longer
- Look at Beta, R, variability, std dev... and most regressor statistics to find correlatory choices
- Public health policy: Identifies where to invest — for example, if smoking rate and chronic disease are major drivers of death, governments might prioritize prevention campaigns.
- Healthcare economics: Helps justify healthcare spending as a cost-effective investment in lowering mortality.
- Academic research: Provides quantitative backing for correlational studies between lifestyle and health outcomes.

Common Examples

- Smoking
- BMI
- Sex
- Exercise time



Research Question:
What factors impact mortality risk in
substantial ways?





Directory Skeleton (Python)

medical-risk-regression/

- |— README.md # overview, setup, how to run [Joseph \(week 1\)](#)
- |— config/ # params, data paths, schema notes [Oliver \(week 1\)](#)
- |— resources/ # raw/ external/ interim/ processed (gitignored) [Joseph \(week 1\)](#)
- |— src/ # data prep, features, model code (packages) [Oliver/Joseph \(week 3\)](#)
- |— scripts/ # make_dataset, train, eval, predict [Joseph \(week 4-5\)](#)
- |— models/ # saved models + metrics artifacts [Oliver \(week 4-5\)](#)
- |— reports/ # figures, tables (coef, CIs) [Joseph \(Week 6\)](#)
- |— docs/ # methodology, data dictionary, results summary [Oliver \(Week 1\)](#)
- |— .gitignore # ignore data/, models/, notebooks checkpoints

Libraries:

pandas
numpy
matplotlib
scipy
statsmodels



Data

CDC BRFSS — behavioral risk factors (smoking, BMI/obesity, alcohol, exercise) —

<https://www.cdc.gov/brfss/index.html> [CDC](#)

NHANES — measured biometrics & labs (height/weight→BMI, smoking, comorbidities) —

<https://www.cdc.gov/nchs/nhanes/index.html> [CDC](#)

NHIS — health status & behaviors (smoking, chronic conditions, demographics) —

<https://www.cdc.gov/nchs/nhis/index.html> [CDC](#)

County Health Rankings — county-level outcomes & determinants (smoking, obesity, premature

death) — <https://www.countyhealthrankings.org/health-data> [County Health Rankings & Roadmaps](#)

WHO Global Health Observatory — international indicators (smoking prevalence, obesity,

mortality) — <https://www.who.int/data/gho> [World Health Organization](#)

MIMIC-IV (PhysioNet) — clinical outcomes incl. in-hospital mortality; rich EHR covariates —

<https://physionet.org/content/mimiciv/>



Data Wrangling

Schema & typing: enforce types (age int, BMI float, smoker $\in$ {never, former, current}); drop impossible values.

Cleaning: standardize, trim outliers (sane BMI/age bounds), de-dup.

Missing data: visualize, NA patterns (drop, fill, or interpolate)

Feature engineering:

- BMI (continuous + bins), age (linear + spline/age²), smoking dummies, interactions (age $\times$ smoking, BMI $\times$ sex).
- Socio-economic controls if available (education, income).



Regression

Outcome

- Binary risk (e.g., mortality flag) → logit/probit
- Continuous risk index/score → OLS with robust SEs

Core specification (Ideas)

- OLS: $\text{Risk}_i = \beta_0 + \beta_1 \cdot \text{Age}_i + \beta_2 \cdot \text{BMI}_i + \beta_3 \cdot \text{Smoker}_i + \gamma^T \cdot X_i + \varepsilon_i$
- Logit: $\Pr(\text{Risk}_i = 1 \mid Z_i) = \Lambda(Z_i)$, with $Z_i = \beta_0 + \beta_1 \cdot \text{Age}_i + \beta_2 \cdot \text{BMI}_i + \beta_3 \cdot \text{Smoker}_i + \gamma^T \cdot X_i$

where $\Lambda(z) = 1 / (1 + e^{(-z)})$ (from logistic CDF)

Covariates (X_i)

- Sex, race/ethnicity, education, income, and region



Regression Interpretation and Diagnostics

Interpretation

- OLS: β_j = partial effect on risk units
- Logit marginal effect at means (MEM): $ME_j = \lambda(\hat{Z}) \cdot \beta_j$, with $\lambda(z) = \Lambda(z) \cdot [1 - \Lambda(z)]$

Diagnostics

- OLS: R^2 , adj- R^2 , heteroskedastic-robust SEs, VIF for multicollinearity
- Logit: AUC (Discrimination), Brier score (MSE of predicted probabilities)

Reporting

- Coefficient table with robust/cluster SEs and 95% CIs
- Marginal effects for Age, BMI, Smoker (report units)
- Goodness-of-fit (AUC or R^2) and calibration chart reference



Potential Problems

- **Bias & confounding:** selection/nonresponse; omitted variables (genetics, access to care); reverse causality (illness $\rightarrow$ weight/smoking).
- **Data quality:** self-reported height/weight/smoking; missing-not-at-random; multicollinearity among SES/health covariates.
- **Design issues:** ignoring survey weights/clusters; overfitting/leakage; class imbalance distorting accuracy.



Some More Applications

Policy targeting: quantify effects of age/BMI/smoking to prioritize prevention (e.g., smoking).

Clinical/insurer ops: risk stratify for outreach; monitor calibration to avoid inequitable decisions.

Geographic planning: map predicted risk by county/tract to allocate resources, depending on datasets



The End